

**Problem 8.27.** Recall that a directed graph is strongly connected if every two nodes are connected by a directed path in each direction. Let

$$STRONGLY-CONNECTED = \{\langle G \rangle \mid G \text{ is a strongly connected graph}\}.$$

Show that *STRONGLY-CONNECTED* is NL-complete.

*Proof.* To show that *STRONGLY-CONNECTED* is NL-complete, we must show that it is in NL and that all NL problems are log space reducible to it.

To prove that *STRONGLY-CONNECTED* is NL-hard, we give a log space reduction from *PATH* to *STRONGLY-CONNECTED*. This reduction converts a directed graph  $G$  and two nodes  $s$  and  $t$  to a directed graph  $G'$  so that  $G$  has directed path from  $s$  to  $t$ , iff  $G'$  is strongly connected.

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